Lecture 3: Basis Vectors, Subspaces, and Products

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🔑 Learning Objectives

By the end of this lecture, you should be able to understand:

- Basis
- Orthonormality
- Subspaces
- Inner product and outer product

📌 Before We Begin

Vectors can exist in both *discrete* and *continuous* forms. To make the concepts easier to grasp, the discrete version of the material will be explained first. Then, the corresponding continuous version and how they relate to each other will be discussed.

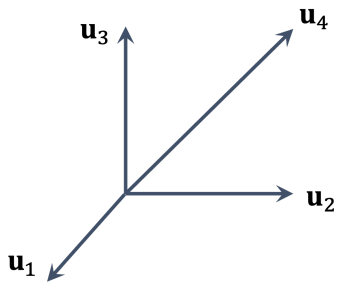
Spanning Set

Let $\mathbf{f} \in \mathbb{U}^N$. It is possible to find a set of N vectors $\{\mathbf{u}_n, n = 1, \ldots, N\}$ such that $\mathbf{f}$ can be represented as:

$$\mathbf{f} = \sum_{n=1}^{N} \alpha_n \mathbf{u}_n,$$

where α_n are scalar coefficients, also known as expansion coefficients.

If all vectors in the space $\mathbb{U}^N$ can be represented as a linear combination of the set $\{\mathbf{u}_n\}$, then this set is called a ***spanning set*** or said to be ***complete***.



The space $\mathbb{U}^N$ is spanned by $\{\mathbf{u}_n\}$, which is written as:

$$\mathbb{U}^N = \text{Span}(\{\mathbf{u}_n, n = 1, \ldots, N\}).$$

📝 Example: Spanning Sets for $\mathbb{E}^3$

$$\mathbf{u}_1 = (1, 0, 0), \quad \mathbf{u}_2 = (0, 1, 0), \quad \mathbf{u}_3 = (0, 0, 1), \quad \mathbf{u}_4 = (0, 1, 1)$$

In [1]:
```python
import numpy as np

# Define a set of four vectors {u_n, n=1,...,4} in E^3
u_1 = np.array([1, 0, 0])
u_2 = np.array([0, 1, 0])
u_3 = np.array([0, 0, 1])
u_4 = np.array([0, 1, 1])

# Print the four vectors
print(f"u_1: {u_1}")
print(f"u_2: {u_2}")
print(f"u_3: {u_3}")
print(f"u_4: {u_4}")
```

```
u_1: [1 0 0]
u_2: [0 1 0]
u_3: [0 0 1]
u_4: [0 1 1]
```

$$\mathbf{u}_1 = (1, 0, 0), \quad \mathbf{u}_2 = (0, 1, 0), \quad \mathbf{u}_3 = (0, 0, 1), \quad \mathbf{u}_4 = (0, 1, 1)$$

Set of Vectors	Description
$\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4\}$	Spanning set, but $\mathbf{u}_4$ is linearly dependent (redundant)
$\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$	Spanning set
$\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_4\}$	Spanning set

Set of Vectors	Description
$\{\mathbf{u}_1, \mathbf{u}_3, \mathbf{u}_4\}$	Spanning set
$\{\mathbf{u}_1, \mathbf{u}_2\}$	Incomplete; does not span $\mathbb{E}^3$
$\{\mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4\}$	Incomplete; does not span $\mathbb{E}^3$

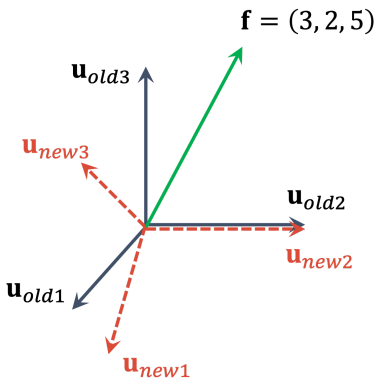
Basis Vectors

Vectors in a ***linearly independent spanning*** set, i.e., a ***minimal spanning*** set

The only solution of $\sum_{n=1}^{N} \alpha_n \mathbf{u}_n = \mathbf{0}$ is $\alpha_1 = \alpha_2 = \cdots = \alpha_N = 0$

$$\begin{aligned}&\text{Number of vectors in a minimal spanning set}\\ &= \text{Number of basis vectors}\\ &= \text{Dimension of the space}\end{aligned}$$

The choice of a basis is not unique.



Example (Continued)

$$\mathbf{u}_1 = (1,0,0), \quad \mathbf{u}_2 = (0,1,0), \quad \mathbf{u}_3 = (0,0,1), \quad \mathbf{u}_4 = (0,1,1)$$

Set of Vectors	Description	Basis?
$\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4\}$	Spanning set, but $\mathbf{u}_4$ is linearly dependent (redundant)	No
$\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$	Minimal spanning set	**Yes**
$\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_4\}$	Minimal spanning set	**Yes**
$\{\mathbf{u}_1, \mathbf{u}_3, \mathbf{u}_4\}$	Minimal spanning set	**Yes**
$\{\mathbf{u}_1, \mathbf{u}_2\}$	Incomplete; does not span $\mathbb{E}^3$	No
	Incomplete; does not span $\mathbb{E}^3$	No

Set of Vectors	Description	Basis?
$\{\mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4\}$		

Standard Basis for $\mathbb{E}^3$

The standard basis for $\mathbb{E}^3$ consists of the following vectors:

$$\mathbf{u}_1 = (1, 0, 0), \quad \mathbf{u}_2 = (0, 1, 0), \quad \mathbf{u}_3 = (0, 0, 1)$$

In [2]:
```
# Standard basis for E^3
u_standard_1 = np.array([1., 0, 0])
u_standard_2 = np.array([0, 1., 0])
u_standard_3 = np.array([0, 0, 1.])

# Expansion coefficients for the standard basis
alpha_standard_1 = 3.
alpha_standard_2 = 2.
alpha_standard_3 = 5.

# Alternate basis for E^3
u_1 = np.array([(2**0.5)/2, 0, -(2**0.5)/2])
u_2 = np.array([0, 1., 0])
u_3 = np.array([(2**0.5)/2, 0, (2**0.5)/2])

# Expansion coefficients for the alternate basis
alpha_1 = -2/(2**0.5)
alpha_2 = 2.
alpha_3 = 8/(2**0.5)
```

In [3]:
```
# Calculate the same vector f using both bases and their expansion coefficients
f_standard  = alpha_standard_1 * u_standard_1 \
    + alpha_standard_2 * u_standard_2 \
    + alpha_standard_3 * u_standard_3
f_alternate = alpha_1 * u_1 \
    + alpha_2 * u_2 \
    + alpha_3 * u_3

# Print vector f
print(f"Vector f calculated using the standard basis and its coefficients:  {f_standard}
print(f"Vector f calculated using the alternate basis and its coefficients: {f_alternate
```

```
Vector f calculated using the standard basis and its coefficients:  [3. 2. 5.]
Vector f calculated using the alternate basis and its coefficients: [3. 2. 5.]
```

Orthonormality

A set of vectors $\{\mathbf{u}_n, n = 1, \ldots, N\}$ is said to be ***orthonormal*** if the vectors are both ***orthogonal*** and ***normalized***.

Orthogonality

For $m \neq n$:

- $\mathbf{u}_m \perp \mathbf{u}_n$ (orthogonal)
- $\theta(\mathbf{u}_m, \mathbf{u}_n) = 90^\circ$
- $\cos[\theta(\mathbf{u}_m, \mathbf{u}_n)] = 0$
- $\langle \mathbf{u}_m, \mathbf{u}_n \rangle = 0$

Normalization

For all m:

- $\|\mathbf{u}_m\| = 1$ (unit vectors)
- $\|\mathbf{u}_m\| = \sqrt{\langle \mathbf{u}_m, \mathbf{u}_m \rangle} = 1$
- $\langle \mathbf{u}_m, \mathbf{u}_m \rangle = 1$

Orthonormality Condition with Kronecker Delta

The orthonormality condition for a set of vectors can be expressed using the **Kronecker delta** δ_{mn} defined as:

$$\delta_{mn} = \begin{cases} 1, & \text{if } m = n, \\ 0, & \text{if } m \neq n. \end{cases}$$

Using the Kronecker delta, the inner product condition for orthonormal vectors can be written as:

$$\langle \mathbf{u}_m, \mathbf{u}_n \rangle = \delta_{mn}.$$

In [4]:
```
# Check the orthonormality of the alternate basis
# Calculate inner products between basis vectors (orthogonality)
inner_product_u1_u2 = np.dot(u_1, u_2)
inner_product_u2_u3 = np.dot(u_2, u_3)
inner_product_u1_u3 = np.dot(u_1, u_3)

# Calculate the L2 norm of basis vectors (normalization)
norm_u1 = np.linalg.norm(u_1)
norm_u2 = np.linalg.norm(u_2)
norm_u3 = np.linalg.norm(u_3)

# Print orthonormality results
print(f"Inner product between u_1 and u_2 (should be 0): {inner_product_u1_u2}")
print(f"Inner product between u_2 and u_3 (should be 0): {inner_product_u2_u3}")
print(f"Inner product between u_1 and u_3 (should be 0): {inner_product_u1_u3}")
print(f"L2 norm of u_1 (should be 1): {norm_u1}")
print(f"L2 norm of u_2 (should be 1): {norm_u2}")
print(f"L2 norm of u_3 (should be 1): {norm_u3}")
```

```
Inner product between u_1 and u_2 (should be 0): 0.0
Inner product between u_2 and u_3 (should be 0): 0.0
Inner product between u_1 and u_3 (should be 0): 0.0
L2 norm of u_1 (should be 1): 1.0
L2 norm of u_2 (should be 1): 1.0
L2 norm of u_3 (should be 1): 1.0
```

Scalar Coefficients/Expansion Coefficients

🧠 *Recall the inner product properties* (in Lecture 2):

- $\langle \mathbf{u}, a\mathbf{v} \rangle = a\langle \mathbf{u}, \mathbf{v} \rangle, \quad a \in \mathbb{C}$
- $\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$

If $\mathbf{u}_n$ are orthonormal basis vectors, the expansion coefficients in the representation $\mathbf{f} = \sum_{n=1}^{N} \alpha_n \mathbf{u}_n$ can be easily determined as:

$$\begin{aligned}\langle \mathbf{u}_m, \mathbf{f} \rangle &= \langle \mathbf{u}_m, \alpha_1 \mathbf{u}_1 \rangle + \langle \mathbf{u}_m, \alpha_2 \mathbf{u}_2 \rangle + \cdots + \langle \mathbf{u}_m, \alpha_N \mathbf{u}_N \rangle \\ &= \alpha_1 \langle \mathbf{u}_m, \mathbf{u}_1 \rangle + \alpha_2 \langle \mathbf{u}_m, \mathbf{u}_2 \rangle + \cdots + \alpha_N \langle \mathbf{u}_m, \mathbf{u}_N \rangle \\ &= \sum_{n=1}^{N} \alpha_n \langle \mathbf{u}_m, \mathbf{u}_n \rangle = \sum_{n=1}^{N} \alpha_n \delta_{mn} = \alpha_m.\end{aligned}$$

In [5]:
```python
# Calculate expansion coefficients for vector f using the alternate basis
alpha_est_1 = np.dot(u_1, f_alternate)
alpha_est_2 = np.dot(u_2, f_alternate)
alpha_est_3 = np.dot(u_3, f_alternate)

# Print results
print(f"Vector f: {f_alternate}")
print(f"Original expansion coefficients:   ({alpha_1}, {alpha_2}, {alpha_3})")
print(f"Calculated expansion coefficients: ({alpha_est_1}, {alpha_est_2}, {alpha_est_3})")
```

```
Vector f: [3. 2. 5.]
Original expansion coefficients:   (-1.414213562373095, 2.0, 5.65685424949238)
Calculated expansion coefficients: (-1.414213562373095, 2.0, 5.656854249492381)
```

❓ Do Hilbert spaces always have finite dimensionality?

🧠 *Recall: Finite-dimensional Euclidean space $\mathbb{E}^N$ is just one example of a Hilbert space.*

Discrete vectors include not only finite-dimensional discrete vectors (Euclidean vectors) but also countably infinite-dimensional discrete vectors (in Lecture 2).

Countably infinite-dimensional Hilbert spaces have a countably infinite number of basis vectors.

The representation of a vector in a Hilbert space is as follows:

Finite-dimensional Hilbert space	Countably infinite-dimensional Hilbert space
$\mathbf{f} = \sum_{n=1}^{N} \alpha_n \mathbf{u}_n$	$\mathbf{f} = \sum_{n=1}^{\infty} \alpha_n \mathbf{u}_n$

⭐ *Hilbert spaces with a countable basis are said to be **separable**.*

Extending to Continuous Bases

Example: Fourier Series

For $f(x) \in \mathbb{L}_2(-\frac{L}{2}, \frac{L}{2})$, the function $f(x)$ can be represented as:

$$f(x) = \frac{1}{\sqrt{L}} \sum_{-\infty}^{\infty} F_n \exp\left(\frac{2\pi i n x}{L}\right) = \sum_{-\infty}^{\infty} F_n \left[\frac{1}{\sqrt{L}} \exp\left(\frac{2\pi i n x}{L}\right)\right],$$

which corresponds to the discrete vector form:

$$\mathbf{f} = \sum_{n=1}^{N} \alpha_n \mathbf{u}_n.$$

The expansion coefficients are given by:

$$F_n = \frac{1}{\sqrt{L}} \int_{-\frac{L}{2}}^{\frac{L}{2}} dx\, f(x) \exp\left(\frac{-2\pi i n x}{L}\right)$$
$$= \int_{-\frac{L}{2}}^{\frac{L}{2}} dx \left[\frac{1}{\sqrt{L}} \exp\left(\frac{2\pi(-i) n x}{L}\right)\right] f(x),$$

which corresponds to:

$$\alpha_n = \langle \mathbf{u}_n, \mathbf{f} \rangle.$$

💡 *Recall: In the discrete form,* $\langle \mathbf{u}, \mathbf{f} \rangle = \sum_{n=1}^{N} u_n^* f_n$, *and in the continuous form,* $\langle \mathbf{u}, \mathbf{f} \rangle = \int dx\, u^*(x)\, f(x)$.

Continuous Basis Vectors

The basis vectors for the continuous case are:

$$u_n(x) = \frac{1}{\sqrt{L}} \exp\left(\frac{2\pi i n x}{L}\right).$$

Orthonormality Condition

The orthonormality condition for the basis vectors is:

$$\langle u_m(x), u_n(x) \rangle = \int_{-\frac{L}{2}}^{\frac{L}{2}} dx\, u_m^*(x)\, u_n(x)$$

$$= \int_{-\frac{L}{2}}^{\frac{L}{2}} dx\, \frac{1}{\sqrt{L}} \exp\Big(\frac{-2\pi i m x}{L}\Big)\, \frac{1}{\sqrt{L}} \exp\Big(\frac{2\pi i n x}{L}\Big)$$

$$= \frac{1}{L} \int_{-\frac{L}{2}}^{\frac{L}{2}} dx \exp\Big(\frac{2\pi i (n-m) x}{L}\Big) = \begin{cases} 1, & \text{if } m = n, \\ 0, & \text{if } m \neq n. \end{cases}$$

This verifies the orthonormality of the basis vectors.

Subspaces

Suppose we have orthogonal vectors $\{\mathbf{u}_n, n = 1, \ldots, N\} \in \mathbb{U}^N$. Then:

$$\begin{gathered} \mathbf{u}_m \perp \mathbf{u}_n, \quad \text{if } m \neq n \\ \mathbf{u}_1 \perp \{\mathbf{u}_n, n = 2, \ldots, N\}, \\ \{\mathbf{u}_m, m = 1, 2\} \perp \{\mathbf{u}_n, n = 3, \ldots, N\}, \\ \ldots, \\ \mathbf{u}_N \perp \{\mathbf{u}_n, n = 1, \ldots, N-1\} \end{gathered}$$

Example

$$\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\} \perp \{\mathbf{u}_n, n = 4, \ldots, N\}$$

The space spanned by $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ and the space spanned by $\{\mathbf{u}_n, n = 4, \ldots, N\}$ are ***subspaces*** of $\mathbb{U}^N$.

Conditions for a Subspace

For a space $\mathbb{S}$ to be a subspace of the Hilbert space $\mathbb{U}$, the following conditions must hold:

- $0_{\mathbb{U}} \in \mathbb{S}$
- $\exists\, \mathbf{u}, \mathbf{v} \in \mathbb{S}, \mathbf{u} + \mathbf{v} \in \mathbb{S}$
- $\exists\, \mathbf{u} \in \mathbb{S}, \alpha \in \mathbb{C}, \alpha \mathbf{u} \in \mathbb{S}$
- The $\mathbb{L}_2$ norm and the inner product are defined.

A subspace inherits the properties of the Hilbert space.

Outer Product

$\mathbf{b} \in \mathbb{E}^M$ and $\mathbf{a} \in \mathbb{E}^N$

Outer product of $\mathbf{b}$ and $\mathbf{a}$ is:

$$\mathbf{b}\mathbf{a}^\dagger = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_M \end{bmatrix} \begin{bmatrix} a_1^* & a_2^* & \dots & a_N^* \end{bmatrix} = \begin{bmatrix} b_1 a_1^* & b_1 a_2^* & \dots & b_1 a_N^* \\ b_2 a_1^* & b_2 a_2^* & \dots & b_2 a_N^* \\ \vdots & \vdots & \ddots & \vdots \\ b_M a_1^* & b_M a_2^* & \dots & b_M a_N^* \end{bmatrix}$$

➡️ Matrix multiplication

$\mathbf{b} \in \mathbb{E}^M$ and $\mathbf{a}, \mathbf{c} \in \mathbb{E}^N$

Action of $\mathbf{b}\mathbf{a}^\dagger$ on some $N \times 1$ vector $\mathbf{c}$:

$$(\mathbf{b}\mathbf{a}^\dagger)\mathbf{c} = \mathbf{b}\mathbf{a}^\dagger\mathbf{c} = \begin{bmatrix} b_1 a_1^* & b_1 a_2^* & \dots & b_1 a_n^* & \dots & b_1 a_N^* \\ b_2 a_1^* & b_2 a_2^* & \dots & b_2 a_n^* & \dots & b_2 a_N^* \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ b_m a_1^* & b_m a_2^* & \dots & b_m a_n^* & \dots & b_m a_N^* \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ b_M a_1^* & b_M a_2^* & \dots & b_M a_n^* & \dots & b_M a_N^* \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \\ \vdots \\ c_N \end{bmatrix}$$

$$= \begin{bmatrix} b_1 a_1^* c_1 + b_1 a_2^* c_2 + \dots + b_1 a_n^* c_n + \dots + b_1 a_N^* c_N \\ b_2 a_1^* c_1 + b_2 a_2^* c_2 + \dots + b_2 a_n^* c_n + \dots + b_2 a_N^* c_N \\ \vdots \\ b_m a_1^* c_1 + b_m a_2^* c_2 + \dots + b_m a_n^* c_n + \dots + b_m a_N^* c_N \\ \vdots \\ b_M a_1^* c_1 + b_M a_2^* c_2 + \dots + b_M a_n^* c_n + \dots + b_M a_N^* c_N \end{bmatrix}$$

The operator $\mathbf{b}\mathbf{a}^\dagger$ maps an $N \times 1$ vector $\mathbf{c}$ to an $M \times 1$ vector.

For m-th element:

$$[\mathbf{b}\mathbf{a}^\dagger\mathbf{c}]_m = b_m a_1^* c_1 + b_m a_2^* c_2 + \dots + b_m a_n^* c_n + \dots + b_m a_N^* c_N$$
$$= \sum_{n=1}^{N} [\mathbf{b}\mathbf{a}^\dagger]_{mn} c_n = \sum_{n=1}^{N} b_m a_n^* c_n = \left[\sum_{n=1}^{N} a_n^* c_n \right] b_m = \langle \mathbf{a}, \mathbf{c} \rangle b_m$$

🧠 *Recall the inner product* (in Lecture 2): $\langle \mathbf{u}, \mathbf{v} \rangle = \sum_{n=1}^{N} u_n^* v_n$

$$\mathbf{b}\mathbf{a}^\dagger\mathbf{c} = \langle \mathbf{a}, \mathbf{c} \rangle \mathbf{b} \quad \text{(Vector form)}$$

The operator $\mathbf{b}\mathbf{a}^\dagger$ acting on $\mathbf{c}$ is the vector $\mathbf{b}$ times the scalar $\langle \mathbf{a}, \mathbf{c} \rangle$.

Because $\mathbf{a}^\dagger \mathbf{c} = \langle \mathbf{a}, \mathbf{c} \rangle$ is sclar, $(\mathbf{b}\mathbf{a}^\dagger)\mathbf{c} = \mathbf{b}\mathbf{a}^\dagger\mathbf{c} = \mathbf{b}(\mathbf{a}^\dagger\mathbf{c}) = (\mathbf{a}^\dagger\mathbf{c})\mathbf{b}$.

The **inner product** is expressed as:

$$\langle \mathbf{u}, \mathbf{v} \rangle = \sum_{n=1}^{N} u_n^* v_n = \mathbf{u}^\dagger \mathbf{v}$$